

Docker Assignment: 4

Docker Multi-stage builds:

The screenshot shows a VS Code editor with two Dockerfiles and a terminal window. The top Dockerfile, 'Dockerfile-multistage', is a multi-stage build. The bottom Dockerfile, 'Dockerfile', is a single-stage build. The terminal shows the directory listing and the successful execution of 'docker build' and 'docker images'.

```
Dockerfile-multistage
1 # ----- Stage 1: Build React App -----
2 FROM node:20-alpine AS build
3
4 WORKDIR /app
5
6 # copy dependency files first (for caching)

Dockerfile
1 #ARG NODE_VERSION=20
2 FROM node:20-alpine
3 WORKDIR /app
4 COPY package*.json ./
5 RUN npm install
6 COPY . .
```

Terminal Output:

```
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> dir

Directory: C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend

Mode                LastWriteTime         Length Name
----                -
d-----          27-07-2026    18:42             build
d-----          28-07-2026    21:52          node_modules
d-----          27-07-2026    18:42             public
d-----          27-07-2026    18:42             src
-a-----          28-07-2026    21:57           141 Dockerfile
-a-----          28-07-2026    21:21        761102 package-lock.json
-a-----          27-07-2026    18:42          660 package.json

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker build -t react-single .
=> => naming to docker.io/library/react-single:latest          0.0s
=> => unpacking to docker.io/library/react-single:latest      18.7s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/jy0ikxx1ln2vojom7190h52q1
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
-----                                -
alpine:latest                        28bd5fe8b56d                      13MB         3.93MB      U
gcr.io/k8s-minikube/kicbase:v0.0.50  b97074569ae9                      1.94GB       545MB
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93 eb4fec00e8ad                      1.94GB       545MB      U
python-backend:v1                    ff9d10cb2d8e                      238MB        60MB        U
react-frontend:latest                 ea60a1d29dd0                      1.29GB       215MB        U
react-single:latest                   6500e2b06f4e                      1.29GB       215MB
```

```
4 WORKDIR /app
5
6 # copy dependency files first (for caching)
7 COPY package*.json ./
8 RUN npm install
9 COPY . .
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker build -f Dockerfile-multistage -t react-multi .

[+] Building 78.6s (19/19) FINISHED

=> [internal] load build definition from Dockerfile-multistage

=> => transferring dockerfile: 875B

=> [internal] load metadata for docker.io/library/nginx:alpine

=> [internal] load metadata for docker.io/library/node:20-alpine

=> [auth] library/nginx:pull token for registry-1.docker.io

=> [auth] library/node:pull token for registry-1.docker.io

=> [internal] load .dockerignore

=> => transferring context: 2B

=> [build 1/6] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293

=> => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293

=> [internal] load build context

=> => transferring context: 3.26MB

=> [stage-1 1/5] FROM docker.io/library/nginx:alpine@sha256:4a73073bd557c65b759505da0378980b61f1be6cbcc3c2c3aeac22d2a470c1752

=> => resolve docker.io/library/nginx:alpine@sha256:4a73073bd557c65b759505da0378980b61f1be6cbcc3c2c3aeac22d2a470c1752

=> [internal] load build context

=> => transferring context: 20.31MB / 20.31MB

=> sha256:390dc935348d8070e695fbaae2a4bb114fb9e69c59f628e7576036ee9d5244c9 1.40kB / 1.40kB

=> sha256:46519e7231d2eb5604df229beb44d59719a489eaa7aca52982535a010b07a9ed 20.31MB / 20.31MB

=> sha256:d0088c891db48b5f526d914bce9e8d889fe1a9d1f08291ae03fe97f871726f38 1.21kB / 1.21kB

=> sha256:46f977ee452f4399c208714afa034868d6056864f8a0cf3c643ab143dd802c80 404B / 404B

=> sha256:62bec68d7c31c4c8a19d812d84da5f7748e54690c037979945b6c5b6c924b142 957B / 957B

=> sha256:1223f016b4e4a2c21f7c49d4837fbfd47a9da6436b511690ca1e582fc2810d59 627B / 627B

=> sha256:3cd534fe98c64d68a1f4f1c83abb8d5c7ecfd7be88e592389929d12e6253da 1.89MB / 1.89MB

View build details: [docker-desktop://dashboard/build/desktop-linux/desktop-linux/kfusxsumys6br48d6a1jdjez](#)

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker images

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	28bd5fe8b56d	13MB	3.93MB	U
gcr.io/k8s-minikube/kicbase:v0.0.50	b97074569ae9	1.94GB	545MB	U
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ff85f95afcd5b5d8d9deb576f3e93	eb4fec00e8ad	1.94GB	545MB	U
python-backend:v1	ff9d10cb2d8e	238MB	60MB	U
react-frontend:latest	ea60a1d29dd0	1.29GB	215MB	U
react-multi:latest	ed7ca3266236	94.8MB	26.5MB	
react-single:latest	6500e2b06f4e	1.29GB	215MB	

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker login

Authenticating with existing credentials... [Username: msaranya]

Info -> To Login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded

7 COPY package*.json ./

1 #ARG NODE_VERSION=20

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker images

react-multi:latest	ed7ca3266236	94.8MB	26.5MB
react-single:latest	6500e2b06f4e	1.29GB	215MB

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker login

Authenticating with existing credentials... [Username: msaranya]

Info -> To Login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker tag react-multi:latest msaranya/react-multi:latest

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker push react-multi:latest

The push refers to repository [docker.io/library/react-multi]

55afa1ecc21d: Waiting

62bec68d7c31: Waiting

3cd534fe98c6: Waiting

d0088c891db4: Waiting

1223f016b4e4: Waiting

390dc935348d: Waiting

1f27c7dcc39d: Waiting

b77fb52e1af1: Waiting

46f977ee452f: Waiting

46519e7231d2: Waiting

ff34b6451a39: Waiting

3cba747b3262: Waiting

504f0d01d1b0: Waiting

push access denied, repository does not exist or may require authorization: failed to authorize: failed to fetch oauth token: Post "https://auth.docker.io/token": push access denied, repository does not exist or may require authorization: server message: insufficient_scope: authorization failed

```
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend> docker push msaranya/react-multi:latest
The push refers to repository [docker.io/msaranya/react-multi]
b77fb52e1af1: Pushed
ff34b6451a39: Pushed
46519e7231d2: Pushed
d0008c891db4: Pushed
46f977ee452f: Pushed
3cba747b3262: Pushed
55afa1ecc21d: Mounted from library/alpine
504f0d01d1b0: Pushed
3cd534fe98c6: Pushed
390dc935348d: Pushed
62bec68d7c31: Pushed
1223f016b4e4: Pushed
1f27c7dcc39d: Pushed
latest: digest: sha256:ed7ca3266236f7aa5faf439f201d66f7cc09a3614cdc058cf22312f720dfc685 size: 856
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\react-frontend\react-frontend>
```

msaranya/react-multi general

hub.docker.com/repository/docker/msaranya/react-multi/general

msaranya
Your personal account

Repositories / react-multi / General

msaranya/react-multi
Last pushed 3 minutes ago · Repository size: 25.3 MB · ☆ 0 · ↓ 0

Add a description
Add a category

Docker commands
To push a new tag to this repository:
docker push msaranya/react-multi:tagname

General Tags Image Management Collaborators Webhooks Settings

Tags
This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	3 minutes

See all

DOCKER SCOUT INACTIVE
Activate

buildcloud
Build with Docker Build Cloud
Accelerate image build times with access to cloud-based builders and shared cache.
Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Repositories | msaranya

hub.docker.com/repositories/msaranya

msaranya
Your personal account

Repositories

All repositories within the msaranya namespace.

Search by repository name All content Create repository

Name	Last Pushed	Contains	Visibility	Scout
msaranya/react-multi	5 minutes ago	IMAGE	Public	Inactive
msaranya/nutritionix-wishlist	over 2 years ago		Public	Inactive
msaranya/nutritionix-service	over 2 years ago	IMAGE	Public	Inactive
msaranya/nutritionix-authentication	over 2 years ago	IMAGE	Public	Inactive
msaranya/userprofile	over 2 years ago	IMAGE	Public	Inactive
msaranya/nutritionix-config	over 2 years ago	IMAGE	Public	Inactive
msaranya/nutritionix-eureka	over 2 years ago	IMAGE	Public	Inactive

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